

Custom silicon programs are difficult to operate

Stratos Silicon • A fictional semiconductor supplier serving AI infrastructure customers

Industry context

AI companies build custom chips for their training and inference workloads. Delivering them requires close coordination across design, testing, validation, and manufacturing.

Why it is challenging

Evidence and actions are spread across five systems. Teams manually reconcile what changed, what is ready, and who needs to approve next.

Business problem

Requirements change throughout development and ramp as workloads and performance targets evolve. A change or yield exception can affect supply, schedule, and customer commitments.

\$203.6B

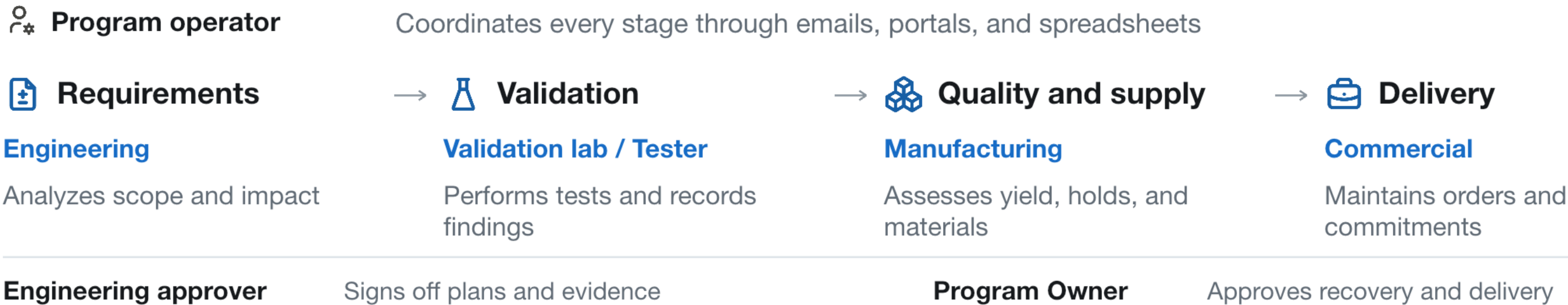
Custom ASIC market by 2030¹

\$10M+

Illustrative annual output value exposed per yield point²

\$2M / business day

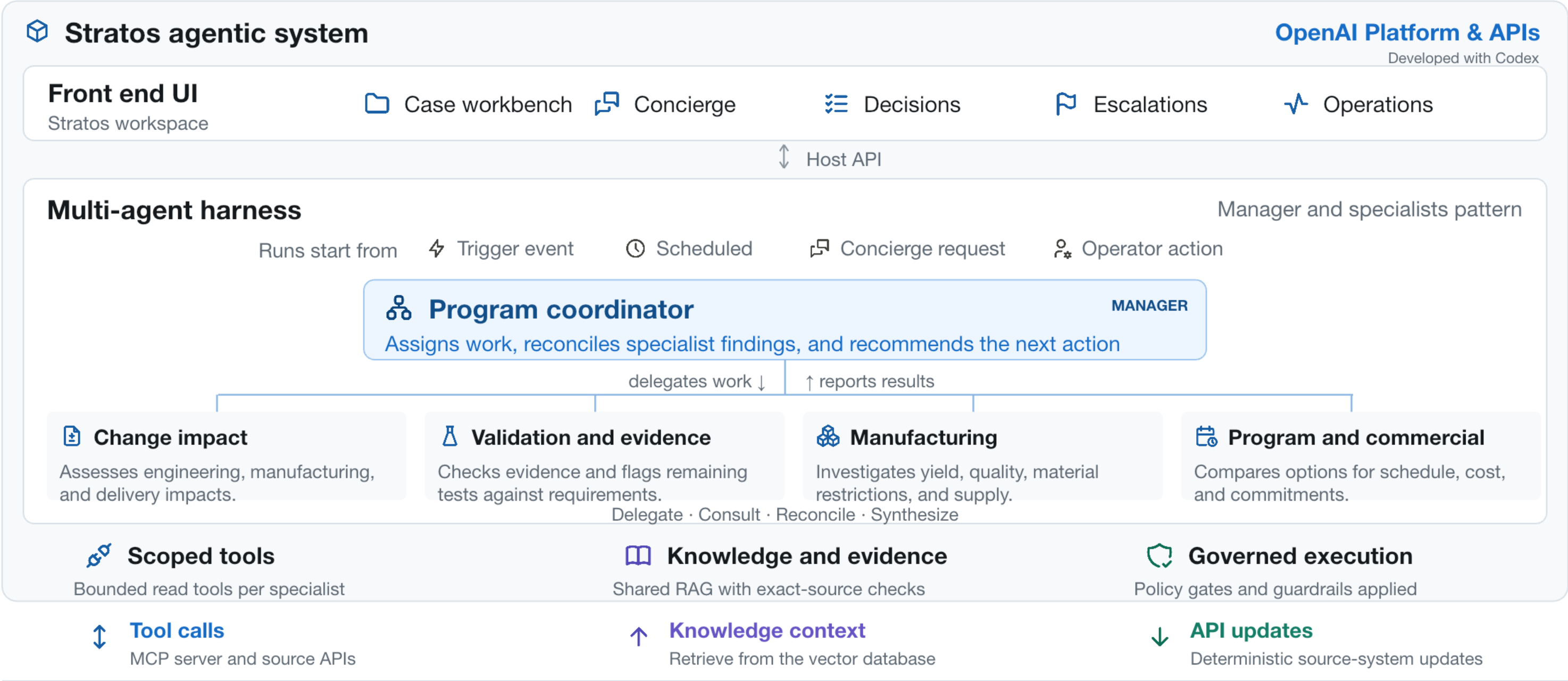
Shipment value exposed to delay²



¹ Gartner, Aug 2026. Industry market, not software TAM. ² Synthetic exposure examples; assumptions in notes.

The solution architecture

One Stratos workspace, a coordinator and four specialists, connected to five source systems



PLM

Engineering records



TestOps

Test evidence and jobs



MES

Quality and materials



ERP

Orders and commitments



ProgramOps

Milestones and tasks

Demo use cases and business value

Two requirement changes and one yield / quality exception, from investigation to verified action

Trigger	→ Investigate	→ Recommend	→ Policy gate	→ Execute	→ Verify
Event, operator, Concierge, or schedule	Specialists retrieve facts from five systems	Options, evidence, and business impact	Within policy: touchless Otherwise: approval	Only the permitted action runs	Read back from the source system
CR-019	Standard requirement change		Within an approved test profile. Reuse matching evidence and route the Validation Operations intake.		Touchless
CR-017	Material requirement change		Find gaps for a broader workload. Propose a validation plan, obtain approval, then schedule and verify the agreed work.		Engineering approval
QE-004	Yield and quality recovery		Assess usable supply and timing. Recommend recovery steps and persist only the approved recovery plan and task.		Program Owner approval

Illustrative value drivers

\$20M

Yield protected

Potential annual output value per percentage point recovered

5 days → 1

Time to decision

Pilot target for an actionable, evidence-backed decision

\$25M

Commitment reliability

Annual shipment value moved out of late-commitment exposure

10 days

Working capital

Less inventory waiting on decisions. Dollar release to validate in pilot.

Design decisions and governance

Models reason. Code calculates and enforces policy. People retain consequential authority.

01 Agent architecture and orchestration

A coordinator delegates to scoped specialists in parallel, then reconciles one recommendation.

03 Tooling and system integration

Typed tools connect all five systems. Source applications remain authoritative.

05 State and execution reliability

Versioned proposals, stale-state checks, idempotency, and independent readback protect execution.

07 Performance, cost, and scale

Use models for ambiguous reasoning and code for routine work. Keep runs bounded and isolated.

02 Grounding and enterprise knowledge

Live APIs supply facts. Specialist RAG and exact-source checks provide eligible context.

04 Governance and human authority

Deterministic policy sets touchless scope. Exact approvals protect engineering, quality, and commitments.

06 Evals, observability, and audit

Test correctness, retrieval, authority, and calculations. Link traces to evidence, decisions, and outcomes.

08 Product UX and operating model

Manual, conversational, and event starts share governed services, with persona-specific evidence and decisions.

Synthetic data and demo identities today. Production authentication and operational controls are the next step.

Production and future expansion

Harden the same architecture, then expand autonomy as measured results allow

01

Connect and secure

Identity, real systems, and data

Enterprise identity

SSO, enterprise groups, RBAC, and least-privilege access to tools, data, and actions.

Real integrations

Connect production PLM, MES, ERP, TestOps, and planning. Preserve source authority.

Security and data governance

Secrets, encryption, isolation, retention, audit logging, and prompt/tool abuse protections.

02

Run reliably at scale

Runtime, routing, and resilience

Dynamic model routing

Choose models by quality, complexity, latency, and cost, with fallback or escalation.

Durable execution

Queues, stateless workers, checkpoints, and bounded retries for concurrent cases.

Execution safety

Keep idempotency, version checks, partial-failure handling, and source readback.

03

Improve and expand

Evaluation and earned autonomy

Observability and evals

Combine OpenAI traces, usage, latency, and cost with workflow outcomes and regression evals.

Continuous improvement

Refine prompts, retrieval, tools, and policies against measured failures and business outcomes.

Measured autonomy

Expand from assistive to touchless only after predefined quality, safety, and value thresholds.

Pilot gate

Measure decision time, execution success, evidence quality, operating cost, and attributable business value.